

MANAGEMENT REVIEW

Management without theatre.

STRATEGY

THE PRE-MORTEM: FAILURE BEFORE IT HAPPENS

Why doubts go unsaid, a thesis that took longer than the worst case, Klein's five steps, what the 30% really measured, what to count afterwards, and a pre-mortem sheet.



48.7%

of the students finished their thesis by the date they had given for the case in which everything went as poorly as it possibly could.

BUEHLER, GRIFFIN & ROSS, 1994

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Most plans are reviewed by the people who wrote them, at the moment they most want them to work. This issue is about a short meeting that starts from the opposite assumption: the plan has already failed, and the team explains why.

Gary Klein described the pre-mortem in Harvard Business Review in 2007. Daniel Kahneman wrote that its main virtue is to make doubts legitimate. In a 1994 study, students finished their thesis later than even their worst-case forecast. The figure most often quoted for the method, 30%, goes back to a 1989 study whose abstract describes something else: how people explain an outcome they take as certain. A 2010 experiment found that the pre-mortem lowered confidence in a plan more than other kinds of critique.

Stiven Janaqi EDITOR · STIVEN CATALYST

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HOW TO READ THIS ISSUE

Figure

Every figure has its source and year at the foot of its page.

Our reading

Where the editors interpret rather than the research, it says so.

Practice

The steps and the card are proposals to try, not research results.

COVER STORY

Permission to doubt

Projects fail at a spectacular rate, Gary Klein wrote in 2007, and one reason is that too many people keep their reservations to themselves while the plan is being made. His remedy turns the usual question around: not what might go wrong, but what did go wrong.

“The main virtue of the premortem is that it legitimizes doubts.”

As a team converges on a decision, Daniel Kahneman wrote, public doubts are gradually suppressed and in the end treated as a sign of flawed loyalty; only supporters keep a voice, and confidence grows. In one of Klein's sessions, a team member who had said nothing in a long kickoff pointed out that an algorithm would not run quickly on the laptops used in the field. The developers had a shortcut they had been reluctant to mention, and it went into the plan.

OUR READING

Teams rarely lack the knowledge of what could fail. They lack a moment in which saying it counts as a contribution, not as disloyalty.

Sources: Gary Klein, Harvard Business Review, 2007; Daniel Kahneman, Farrar, Straus and Giroux, 2011 (via Bloomberg, excerpt of 24 October 2011)

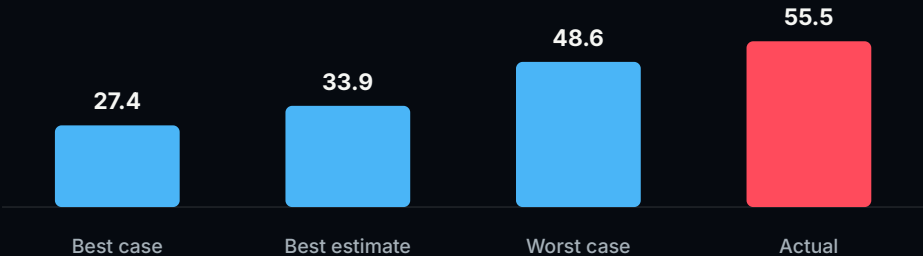
Kahneman's words are quoted from an excerpt of Thinking, Fast and Slow that Bloomberg published in 2011. The laptop case is Klein's own account; we could not check it independently.

THE NUMBERS

Later than the worst case

In 1994 Roger Buehler, Dale Griffin and Michael Ross asked 37 psychology students at the University of Waterloo, in the last term of their honours thesis, when they would hand it in: their best estimate, and the dates if everything went as well, or as badly, as possible.

DAYS TO FINISH THE THESIS, AVERAGE OF 33 STUDENTS, 1994



Only 29.7% finished by their best estimate, 10.8% by the best-case date, 48.7% by the worst-case date. The worst-case forecasts were less biased but no more accurate: they missed by 23.2 days on average, the best estimates by 22.6.

OUR READING

Imagining the worst case still starts from the plan. The pre-mortem starts from the failure and asks how it came about.

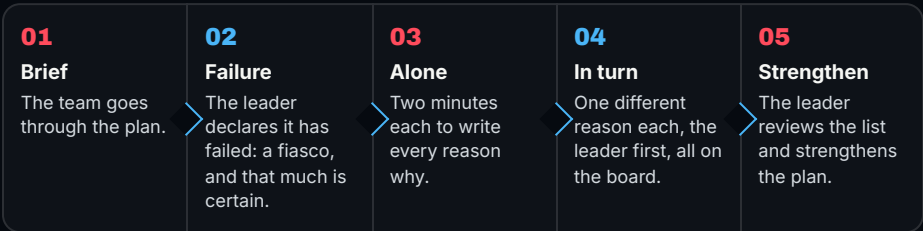
Source: Roger Buehler, Dale Griffin & Michael Ross, Journal of Personality and Social Psychology, 1994

Averages of the 33 students who had finished when the records closed. The gap between the worst case and the actual time was not statistically significant. One group, one task.

THE MODEL

Five steps, half an hour

Klein's pre-mortem comes at the start of a project, once the team knows the plan. In 2021 he wrote that it can take as little as 20 to 30 minutes.



Klein later added two minutes in which each person writes actions they can take, and a rating of each problem for likelihood, impact and ease of prevention. Kahneman places the session where an organisation has almost reached an important decision but has not yet committed itself.

OUR READING

Timing matters as much as method: before the decision, a doubt is input; after it, the same doubt sounds like opposition.

Sources: Gary Klein, Harvard Business Review, 2007; Gary Klein, Psychology Today, 2021; Daniel Kahneman, Farrar, Straus and Giroux, 2011 (via Bloomberg, excerpt of 24 October 2011)

The steps follow Klein (2007; 2021); in 2007 the round began with the project manager, in 2021 with the team leader. The grouping into five is the editors'.

More in the essay

High-volume days: the standard under pressure

WHAT THE RESEARCH SAYS

What the 30% really measured

Klein cites a 1989 study by Deborah Mitchell, Jay Russo and Nancy Pennington: imagining that an event has already occurred, he wrote, raises the ability to correctly identify reasons for future outcomes by 30%. The study's abstract describes something narrower.

01 Mitchell et al., 1989

Varied whether an event lay in the future or the past, and whether it was certain. Time had little effect; certainty mattered: explanations of sure events were longer, with more episodic reasons. No measure of correct reasons is reported.

02 Veinott, Klein & Wiggins, 2010

178 students reviewed a university's plan for an H1N1 outbreak. Pros and cons, cons only and the pre-mortem all lowered confidence more than the baseline; the pre-mortem most.

03 Gallop, Willy & Bischoff, 2016

Teams of experienced programme managers and engineers had 45 minutes on one torpedo case. As judged by an expert panel, pre-mortem teams found more quality risks and plan changes than brainstorming teams.

OUR READING

The studies share a frame, not a number: the failure is taken as certain. That is the part a team can keep.

Sources: Gary Klein, Harvard Business Review, 2007; Deborah J. Mitchell, J. Edward Russo & Nancy Pennington, Journal of Behavioral Decision Making, 1989 (via Crossref, abstract); Beth Veinott, Gary Klein & Sterling Wiggins, Proceedings of the 7th ISCRAM Conference, Seattle, 2010 (via Gary Klein, Psychology Today, 2021); David Gallop, Chris Willy & John Bischoff, Journal of Enterprise Transformation, 2016 (via ResearchGate, abstract)

We could not read the full 1989 paper and give no figure from it. The later studies used one scenario each; lower confidence is not yet a better plan, and we found no study of real project results.

HOW IT IS MEASURED

Did the plan change?

A pre-mortem is not judged by the length of its list. In Klein's account, it pays off as a stronger plan and as a team that picks up early signs of trouble once the work is under way. Four counts show whether that happened.

1

Voices

How many people gave at least one reason? A list written by two people is theirs, not the team's.

2

New reasons

How many had not come up in the kickoff or the risk list?

3

Changes

How many led to a change in the plan, with an owner and a date?

4

Early signs

For the top reasons: which signal shows early that it is happening, and who watches it?

HYPOTHETICAL EXAMPLE, A PRE-MORTEM BEFORE A PEAK WEEK IN A WAREHOUSE

Voices 9 in the room, 8 gave a reason

Reasons 23 on the board, 7 never raised before

Changes 4 to the plan, 2 warning signs with an owner

The line that counts is the last one: what the session changed. The numbers are invented.

Sources: Gary Klein, Harvard Business Review, 2007; Daniel Kahneman, Farrar, Straus and Giroux, 2011 (via Bloomberg, excerpt of 24 October 2011)

The counts and the example are the editors', drawn from the benefits Klein lists. Kahneman warns that the pre-mortem is no panacea and gives no complete protection against nasty surprises.

TOOL OF THE ISSUE

The pre-mortem sheet

Run it before the plan is approved, not after. Read the second line aloud, give everyone two minutes alone, and take the top reasons to the 5 Whys sheet to reach a cause you can change.

01 Plan and decision what we are about to commit to, and by when

02 The failure "It is [date]. We carried out the plan as written. It failed."

03 Why it failed two minutes alone, then one reason each, in turn

04 Top three reasons likelihood, impact, how easy to prevent

05 Changes to the plan what changes, who owns it, by when

06 Early warning signs what we will see first, and who watches for it

Sources: Gary Klein, Harvard Business Review, 2007; Gary Klein, Psychology Today, 2021; Daniel Kahneman, Farrar, Straus and Giroux, 2011 (via Bloomberg, excerpt of 24 October 2011)

A practice proposed by the editors, after Klein (2007; 2021) and Kahneman (2011).

Open the tool: 5 Whys

stivencatalyst.com/tools/five-whys/

SOURCES AND METHOD

Every figure has a source.

The figures in this issue come from the sources below. The year shows how recent each one is.

Performing a Project Premortem

Gary Klein, Harvard Business Review, 2007
<https://hbr.org/2007/09/performing-a-project-premortem>

Thinking, Fast and Slow

Daniel Kahneman, Farrar, Straus and Giroux, 2011 · via Bloomberg, excerpt of 24 October 2011

Exploring the “planning fallacy”: Why people underestimate their task completion times

Roger Buehler, Dale Griffin & Michael Ross, Journal of Personality and Social Psychology, 1994
<https://doi.org/10.1037/0022-3514.67.3.366>

The Pre-Mortem Method

Gary Klein, Psychology Today, 2021
<https://www.psychologytoday.com/us/blog/seeing-what-others-dont/202101/the-pre-mortem-method>

Back to the future: Temporal perspective in the explanation of events

Deborah J. Mitchell, J. Edward Russo & Nancy Pennington, Journal of Behavioral Decision Making, 1989 · via Crossref, abstract
<https://doi.org/10.1002/bdm.3960020103>

Evaluating the Effectiveness of the PreMortem Technique on Plan Confidence

Beth Veinott, Gary Klein & Sterling Wiggins, Proceedings of the 7th ISCRAM Conference, Seattle, 2010 · via Gary Klein, Psychology Today, 2021
https://idl.iscram.org/files/veinott/2010/1049_Veinott_et al2010.pdf

How to catch a black swan: Measuring the benefits of the premortem technique for risk identification

David Gallop, Chris Willy & John Bischoff, Journal of Enterprise Transformation, 2016 · via ResearchGate, abstract
<https://doi.org/10.1080/19488289.2016.1240118>

EDITORIAL METHOD

Each figure was checked for its year, its publisher and what exactly it measures. Where the publisher's page could not be opened, the figure was checked against independent summaries and is marked “via”. The editors' interpretation is marked “Our reading”. Figures that could not be confirmed are not in the issue.



MANAGEMENT REVIEW

Management without theatre.

Every issue, one management question, checked against the best research.

ALL ISSUES

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